Odd Pell helpers and saturated carriers

Research notes for ProveIt

25 October 2026

Abstract

We give eight explicitly specified fixed polynomials whose zero nonexistence, in their stated domains, is equivalent to the syntactic consistency of ZFC at the retained ordinary compiler calibration. Three small-coefficient direct quartics use 307, 308 and 310 unrestricted integer variables, with explicit support and coefficient-height tradeoffs. Root absorption lowers the fourteen-integer degree from 108 to 88 at the same 119-operation cost, and improves the thirteen-integer degree/cost pair from 274/129 to 156/127. A separate eleven-integer equation improves the degree/cost pair from 654/117 to 648/115. Two ten-positive equations reduce the supplied product-square evaluator costs by three operations, to 81 and 82, at increased degrees 750 and 560.

The reductions use signed odd helper quotients, exact integral projections, parity separation within common square classes, unit-product conjunctions and saturated carriers. Complete equations, coefficient payloads, ordinary proofs, source pins, independent audits and offline replay accompany the report. These are arithmetic equivalences, not a proof of consistency, a global optimality theorem or a new end-to-end proof-assistant certificate.

Construction	Domain	Variables	Degree	Gates
Odd helper minimum	$\mathbb{Z}$	307	4	—
Odd helper balanced	$\mathbb{Z}$	308	4	—
Odd helper low height	$\mathbb{Z}$	310	4	—
Two root carrier	$\mathbb{Z}$	14	88	119
Parity separated roots	$\mathbb{Z}$	13	156	127
Amplified product carrier	$\mathbb{Z}$	11	648	115
Doubled Pell square	$\mathbb{Z}_{>0}$	10	750	81
Doubled Pell mixed	$\mathbb{Z}_{>0}$	10	560	82

For the direct quartics, the three support/height pairs are respectively $22,532/394,412$, $5,364/195,962$ and $5,612/128,446$. The final point improves the 324-integer comparator's variable count and support at equal coefficient height, but uses six more residuals. Three additional eleven-integer siblings are retained only as audited, dominated comparisons.

Contents

1 Scope and the exact logical claim **3**

2 The pinned comparison frontier **3**

3 Fixed numerals and the inherited source boundary **4**

4 Direct quartics from an odd helper and cofinal compositions **5**

5 Absorbing a positivity square with fourteen integers **8**

6 Parity separated roots with thirteen integers **10**

7 Canonical saturation of the doubled Pell product square **13**

8 Eleven integers with an amplified product carrier **15**

9 What is checked and how to reproduce it **18**

10 Provenance and source guide **19**

1 Scope and the exact logical claim

The objective is an explicitly specified fixed polynomial with integer coefficients whose zero existence, in the domain stated for the construction, is equivalent to existence of a finite syntactic refutation of ZFC. Writing D for that domain, the ordinary arithmetic equivalence has the form

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The equivalence does not prove either side. The fixed recursive presentation and proof coding are those of the inherited source calibration. No consistency assumption is used to choose coefficients or establish the zero-set reductions.

The new proofs are ordinary unbounded mathematics relative to the retained source theorems. Neither a successful program replay nor source checksums supply a fresh end-to-end Lean, Rocq, or object-language ZFC certificate. The inherited compiler, Pell and formal-system calibration interfaces remain substantive mathematical dependencies. Exact finite computation checks the new transcription, identities, degree certificates and data integrity; it does not replace those unbounded hypotheses.

The domains $\mathbb{N}$, $\mathbb{Z}_{>0}$ and $\mathbb{Z}$ are distinct comparison classes. Each construction states its domain explicitly. A named polynomial register is an abbreviation to be substituted, not an additional existential unknown. Proof-only divisions, Pell sequences and radicals are not operations of the supplied division-free equations.

Circuit cost counts live binary additions, subtractions and multiplications; a square or multiplication by a fixed numeral is one multiplication. An exact degree is distinguished from a syntactic DAG upper bound. Fixed coefficient bit length, variable count, total degree, circuit cost, support and coefficient height are separate criteria. A gain in one criterion need not dominate another construction.

2 The pinned comparison frontier

The comparison is to a bounded repository snapshot, not to an asserted global minimum. The inspected *ProveIt* main head at 04:24 UTC on 25 October 2026 was `b30ddde0aab0`. The last research commit in the initial comparison pass was `e915fd77a49a`, at 04:25:16 UTC. Exact immutable file URLs, Git blob hashes and SHA-256 hashes are retained in `baseline/DOWNLOAD_MANIFEST.json`.

Seven newly placed source packets, numbered 127–133, materially change the comparison with the preceding delivery. In particular, source 131 already contains a complete 324-integer quartic. The independently replayed 329-integer research point is therefore not the smallest materialized comparison in this snapshot. Source 132 is the preceding Signed Caps and Free Scales delivery, not a new result of this iteration.

Direct quartic	Variables	Rows	Degree	Terms	Height
Source 131, unrestricted	324	208	4	5,808	128,446
Earlier accepted refined product, unrestricted	329	216	4	5,728	128,446
Source 131, unrestricted	338	222	4	4,493	128,446
Source 129, nonnegative	233	182	4	50,762	88,674,690
Source 129, sparse nonnegative	234	183	4	9,019	87,741,134

The baseline audit independently re-expands every coefficient of these five quartics, checks their complete variable supports and recomputes their metrics. This is a data-level check. It does not newly prove the authors' unbounded source calibrations. The earlier accepted 329-integer branch has a retained upstream root/peer replay; the newly placed numbered packets retain their author proofs without being represented as newly integrated formal proofs.

For ten strictly positive variables, source 133 already gives operation/degree pairs 84/436 and 85/328. Both use Pell doubling and a product-square equation. They supersede 86/466 on the selected two-metric

frontier, so cubic normalization alone cannot be counted as a new improvement. The earlier gap-fusion pairs are 89/288, 91/283, 92/278, 94/262, 96/258 and 97/254; source 130 also gives 121/246. The last degree depends on exact cancellation, not its syntactic upper bound of 288. Lower-cost, higher-degree points and higher-cost, lower-degree points remain distinct tradeoffs.

For eleven unrestricted integers, selected prior operation/degree pairs are

$$109/924, \quad 110/840, \quad 112/810, \quad 117/654, \quad 118/488, \quad 119/464,$$

$$123/428, \quad 125/416, \quad 127/376, \quad 131/364, \quad 132/352.$$

The 112/810 and 117/654 points are from newly placed source 131. This combined list is more demanding than comparison only with the preceding packet. Each point retains its own complete fixed-coefficient binding. In particular, a mathematically useful sibling need not be a new Pareto point. The preceding fifteen-integer degree-44 result is a separate variable-count tradeoff and is not superseded merely by changing an eleven-integer circuit.

The direct source architecture is also moving. The fused ninth-factor theorem already has a complete 475-coordinate, 364-row source; the support-pair exchange theorem is likewise credited as an inherited reduction. Neither its prepared projection code nor a source-level guard count alone is treated as a measured final quartic. New endpoint claims require the literal final equation and its complete expansion.

A late comparison check found `a6a4cb5a7185`, which accepts the centered 327-integer quartic with 216 residuals, $5,718$ terms, degree 4 and height $128,446$. Its exact upstream review and metrics are retained under `baseline/late/`. This improves the accepted 329 research point, while the newly placed 324-integer source remains the smaller variable-count comparator. The new 327 expansion was not independently recollected by the initial five-quartic baseline audit.

The same late check confirms that the nine-positive 413-operation, degree-$1,316,546$ page is complete, rather than still preparatory. It remains a different variable-count and degree regime from the ten-positive constructions here. Exact late-review URLs and hashes are retained alongside the original comparison snapshot instead of silently changing its provenance.

The retained *Coprime Projection* predecessor also contains a fourteen-integer 108/119 degree/cost point and a thirteen-integer 274/129 point. Their actual literal graphs were freshly recounted, including liveness: respectively $69+30+20 = 119$ and $76+31+22 = 129$ multiplications, additions and subtractions. Their exact-degree proofs remain the retained audited historical claims. The source paths and hashes are recorded in `baseline/HISTORICAL_LOWDEGREE_CHECK.json`; the comparison uses the original `equation.json` profiles, not their differently scheduled siblings.

3 Fixed numerals and the inherited source boundary

The two compact coefficient architectures are not interchangeable. The unrestricted-integer branches retain the complete original-label source34 tuple, including

$$\beta = 2^{1064921} - 1, \qquad A_c = 2^{1113}, \qquad B_c = 2^{666104}.$$

Their high-mask port is $M_F = M_F^{\text{native}} + \beta$ and is already shifted. The eleven-integer checker reconstructs the complete tuple from its finite source recipe, in addition to authenticating every payload byte.

The ten-positive branch instead retains source 133's complete directly indexed BFS943 compact44 tuple. Here $\rho = 2^{67}$, $B = \rho^{8807} = 2^{590069}$ and $\beta = B - 1$. The source-specific initializer, tile ordering and mask convention are fixed jointly with K_0, M_C, M_F. Its full constant specification contains the literal seed, finite indexed tile table and all arithmetic recipes; its indexed native proof is also retained. Merely replacing one coefficient with the same-named port from another family is not an authorized mathematical transformation.

Complete tuple	β bits	K_0 bits	M_C bits	M_F bits
Original-label source34	1,064,921	1,065,043	1,064,921	1,064,922
Directly indexed BFS943	590,069	592,884	590,069	590,070

The two source34 cone coefficients additionally have $1,114$ and $666,105$ bits. These sizes are separate from arithmetic-operation counts. The delivered payloads are gzip containers of unsigned little-endian integers, with raw-byte hashes and bit lengths. For three positive ports the gzip headers differ from the newly placed source container, but every decompressed coefficient byte matches its pinned numeral exactly; the repacking record makes this explicit.

The direct quartics use their complete sparse integer expansions instead of large native coefficient ports. Their semantic chain retains the fixed finite calculus, square-shell encoding, factorial and reciprocal-CRT arithmetic, derived-label bounds, Horn flow and Pell-cap extraction. The fused ninth factor and support-pair exchanges are credited source results. The new cap and exact integral projection steps do not by themselves prove the calculus-to-ZFC calibration.

For every branch the ordinary proof has the same logical structure:

$$\begin{aligned} &\exists \text{ zero of the displayed fixed polynomial} \\ &\quad\Longleftrightarrow\ \exists \text{ accepted finite source certificate} \\ &\quad\Longleftrightarrow\ \exists \text{ finite syntactic ZFC refutation.} \end{aligned}$$

The complete preceding release under `prior/wave16/` retains its coherent transitive source and licensing closure. Current source statements and source pins are also copied beside the constructions. This preserves the inherited mathematical assumptions; it does not assert a fresh reconstruction of the entire formal-system and compiler chain in a theorem-prover kernel.

4 Direct quartics from an odd helper and cofinal compositions

4.1 Three measured endpoints

The smallest-variable endpoint of this branch is the literal equation

$$P_{307}(z_0,\dots,z_{306}) = \sum_{j=0}^{210} R_j(z_0,\dots,z_{306})^2 = 0, \qquad z_i \in \mathbb{Z}.$$

Every quadratic R_j and its ordered coordinates are listed in `direct/centered307/equation.json`; its complete collected quartic is `expanded_polynomial.json` in the same directory. It has 22,532 nonzero monomials and coefficient height 394,412. Two useful siblings are retained rather than hidden by a single-number comparison:

Unknowns	Residuals	Degree	Monomials	Height
307	211	4	22,532	394,412
308	212	4	5,364	195,962
310	214	4	5,612	128,446

The incoming 324-integer endpoint has 208 residuals, 5,808 terms and height 128,446. Thus the 310 endpoint improves variables and support at equal height, but uses six more residuals. The 307 endpoint prioritizes variable count, not support or height. These are comparisons within the inspected direct-calculus small-coefficient architecture, not globally smallest quartics. Other retained architectures have many fewer variables and enormous fixed coefficients.

4.2 The exact new helper simplification

Write W for the even square frame, A for its signed Pell parameter, (U, C) for the single cap, and (H, J) for its auxiliary Pell pair. Retain the quadratic cap and primary-only pooled congruence, but replace

$$J = C^2(e+1) \quad \text{by} \quad J = C^2(2e+1), \qquad e \in \mathbb{Z}.$$

This removes the remaining comparison-base sign graph. The full proof does not assert that its signed comparison base becomes positive.

The cap and nonzero target exclude $|C| = 1$: otherwise all primary ordinates would vanish and a strictly positive sum of small index squares would be divisible by $A - 1$. Hence $|C| > 1$. The odd quotient gives $|J| \geq C^2 > 1$, so Pell classification gives

$$|H| \geq 2|A|^2 - 1 > |A - 1|.$$

If the comparison constructor $D = 1 + 4J(d+1) = A + Hf$ had $d + 1 = 0$, then $H \mid A - 1$, a contradiction. Therefore $|D| \geq 4|J| - 1 \geq 15$. The existing index-residue proof works at negative D by polynomial Pell base transfer and independent coordinate signs. It recovers $|C| = Y_W(|A|)$, then every canonical signed-index primary ordinate.

For completeness choose $U = X_W(A)$, $C = Y_W(A)$ at positive A. Since W is even, U is odd and C is even. Set

$$H = X_C(U) = X_{WC}(A), \qquad J = CY_C(U) = Y_{WC}(A).$$

For odd U, the doubling identity $Y_{2n}(U) = 2X_n(U)Y_n(U)$ and parity at odd indices imply $v_2(Y_n(U)) = v_2(n)$. The expansion $Y_C(U) = CU^{C-1} + (U^2 - 1)T$, with $U^2 - 1 = (A^2 - 1)C^2$, also proves $C \mid Y_C(U)$. Thus J/C^2 is an odd integer. Also H is odd and coprime to $4J$, so CRT supplies a large positive $D \equiv A \pmod{H}$, $D \equiv 1 \pmod{4J}$. Its index-W pair supplies the complete helper. Soundness restores the old positive helper by this fresh construction, not by an invalid whole-tuple sign reflection.

4.3 Cofinal and arithmetic composition

The materialized source also uses the explicitly credited upstream ingredients

$$t = 16 + h_0^2, \quad y = V_0(V_0 - 1), \quad A^2 - ((W+1)^2 - 1)[W(2z+1)]^2 = 1,$$

and the self-sized radix

$$F = t!, \quad V_0 = (F^2+1)^t, \quad B = \binom{F^2+t}{t}, \quad N = B^K, \quad q = (BH_0)^K, \quad H_0 = 1 + \sum_{i=1}^{9} a_i.$$

These powers and factorials describe recovered meanings; they are not operations left in the final polynomial. All 18 powers have literal quadratic Pell lowerings, 15 independent indices and one derived index. The modified frame contains all 38 distinct actual power atoms, including the new base BH_0.

The proof first obtains $q \geq 2$ from retained scalar signs, then orients the t pair and recovers V_0 and $F \geq 1$. The consecutive-product modulus makes $y \geq F + 1$. Positive denominator powers justify three product guards in place of six separate scalar signs. The unchanged positive x anchor is retained. Successor output residues modulo $q \geq 3$ exclude negative encoded indices; the strict eight-factor product excludes zero factors. The ninth zero endpoint is excluded by $Bv_{27} = F^2 + 1$ with $0 < F^2 + 1 < B$. Only after these steps is H_0 used as a positive radix factor. The ordinary binomial, factorial, prime-power CRT and Horn-flow proof then applies with its fully stated new radix bounds.

Completeness starts with a finite refutation, pads its finite proof tree to a sufficiently large square-offset horizon, and then sets $F = t!$. The chosen y is divisible by F and exceeds every proof accumulator, so the beta CRT code exists. Choose the independent representatives before forming H_0 and q. Construct the frame and private Pell witnesses last. This order has no fixed-point or future-modulus circularity.

4.4 Logical and validation boundary

The three charts are not claimed equal as polynomials. They are existentially equivalent by explicit integer coordinate restoration. Write v_i for the unprojected body numbering and $a = v_{16}$, $b = v_{17}$, $c = v_{18}$. The quadratic-first policy, relative to the affine-first 310-coordinate policy, deletes nine additional wires by the exact restorations

$$\begin{aligned}
v_{55} &= v_{54}^2, \\
v_{66} &= 93 + 30a + 32b + 14v_{29} + (24 + 8a + 8b + 4v_{29})^2, \\
v_{69} &= 21 + 8b + 6c + 2v_{35} + v_{67}^2, \\
v_{74} &= (3 + a + 2c + 2v_{73})^2, \\
v_{76} &= v_{75}^2, \\
v_{82} &= (4 + 2b + 2v_{35})^2, \\
v_{98} &= (7 + 2v_{97})^2, \\
v_{106} &= 51 + 3c + 16v_{102} + 8v_{104} + (13 + c + 4v_{102} + 2v_{104})^2, \\
v_{109} &= (9 + 2a + 2v_{29} + 2v_{108})^2.
\end{aligned}$$

It instead retains the seven wires $v_{56}, v_{67}, v_{70}, v_{77}, v_{83}, v_{99}, v_{110}$ that the other order deletes. The net saving is therefore two coordinates and two rows, giving 308, not nine. Each deletion solves a whole current row with unit coefficient; its target occurs only linearly in the remaining current rows when its restoration is quadratic. The full ordered certificate verifies that all substituted rows remain quadratic.

Raising the allowed affine restoration support from 16 to 32 then removes one further wire, with the explicit affine restoration

$$\begin{aligned}
v_{171} = 48 &+ 9a + 9b + 4v_{29} + 2v_{34} + 4v_{37} + 2v_{42} + 2v_{46} + 4v_{49} + 2v_{53} + 2v_{58} \\
&+ 4v_{63} + 2v_{64} + 2v_{72} + 2v_{79} + 4v_{112} + 2v_{113}.
\end{aligned}$$

This gives 307 and causes the larger support cost. Forward, compute these wires and forget them after substitution. Reverse, restore every deleted wire in reverse certificate order, set the actual unconstrained wires to 0, and rebuild the original private Horn witnesses. Both maps are integral and preserve zero existence. The complete certificates also contain every earlier common elimination, not only the distinguishing steps displayed here.

Finally each centering chart is the bijection $v_i = z_i + s_i$ with its full integer shift list supplied in `translation_certificate.json`; the inverse is $z_i = v_i - s_i$. The recorded finite coordinate-descent search changes coefficients, not the zero-existence theorem. No optimality of this elimination order or centering policy is asserted.

The full ordinary proof is `direct/PROOF.md`, with the native arithmetic, square-shell coding, calculus and named ZF/ZFC calibration sources retained locally. It gives, in ZFC,

$$\exists z \in \mathbb{Z}^n \; P_n(z) = 0 \quad \Longleftrightarrow \quad \text{a finite ZFC refutation exists.}$$

Negation is the desired equivalence with Con(ZFC). There is no claim to prove consistency or to supply a new proof-assistant kernel certificate. The independent checker verifies all actual power lowerings, the full frame and pool, 11 successor residue ideals, Horn identities and private scopes, every monic restoration, integer translations and every collected quartic coefficient. Eight altered-artifact controls and finite signed-helper diagnostics supplement, but do not replace, the universal proof. Normal and optimized Python rebuilds are byte-identical.

The mathematical novelty is the exact odd-helper composition and its measured endpoints. The odd-quotient parity trick itself, the cofinal restrictions, self-sized radix, and support-product principle have explicit prior credits. No global minimality or literature-wide historical-priority claim is made.

5 Absorbing a positivity square with fourteen integers

The polynomial below uses fourteen unrestricted integer variables, has exact ordinary total degree 88, and a division-free evaluator with 119 binary operations (67 multiplications, 27 additions and 25 subtractions). It is equivalent by explicit two-way maps on every zero to the pinned fifteen-variable free-scale degree-44 source. Consequently its zero nonexistence is equivalent in ordinary ZFC to syntactic Con(ZFC) under that unchanged compiler/calibration theorem. This is neither a proof of consistency nor a new end-to-end formal-kernel certificate.

5.1 The complete fixed equation

All unknowns are unrestricted integers:

$$j, C, z, h, r, v, w, T, c, n, f, y, p_2, p_3.$$

The six fixed numerals $\beta, K_0, M_C, M_F, A_c, B_c$ are exactly the authenticated source34 payloads, with M_F already shifted. The remaining names are polynomial abbreviations, not extra witnesses:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad Q = \Delta(f^2 - 1), \quad V = f^2(cT - R) - c, \\
M = 2q^3, \quad U = MH, \quad K_{\mathrm{lo}} = X(2X - 3c) - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(Y - q^3), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_{\mathrm{ud}} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2(4(Y\eta)^2 + (kq^3)^2), \\
G = B_{\mathrm{ud}} - 1 - p_2^2 - p_3(p_3 + 1).
\end{gathered}$$

Set

$$\begin{aligned}
C_1 &= (Z + Uk)^2 + 4U(LZ - U), \\
B &= f^2 - \Delta I^2 - 1, \\
C_2 &= V^2 - (Q - 1)y(2V + y) - 1, \\
N &= (C_1 - U^2G)^2 - 16U^4G(1 + L(L + k)).
\end{aligned}$$

The single equation is

$$\boxed{P = (N + 1)(B + 1)(C_2 + 1) - 1 = 0.}$$

The evaluator shares U^2G, q^3 and the already-counted registers. No division, radical, inequality or variable exponent appears in its literal DAG.

5.2 An exact, unsquared conjunction

For integers A, B with $A, B \neq -2$, the equation $(A+1)(B+1)(C+1) = 1$ is equivalent to $A = B = C = 0$: all three factors must be units, and the first two cannot be -1. Here N is a square modulo 4, hence $N \neq -2$. Write $A_0 = a+2$, so $\Delta = A_0^2 - 1$. If A_0 is odd, $B \equiv f^2 - 1 \pmod 4$; if A_0 is even, $B \equiv f^2 + I^2 - 1$

(mod 4). Neither can be 2 (mod 4), so $B \neq -2$. Thus $P = 0$ is exactly $N = B = C_2 = 0$, with no nonnegativity assumption on N. For comparison, the general construction $\prod_i (3R_i + 1) = 1$ also encodes $R_i = 0$ over integers; the present residue exclusions avoid those extra scalar multiplications. No novelty is claimed for that general unit-product device.

5.3 Separation and extraction before invoking the source

Since $\beta = 2^{1064921} - 1$ is odd and greater than one, q is even and nonzero. Also H is odd and nonzero. Thus $U \neq 0$ on every integer assignment, and $|X| = |qW| \geq 4|q| \geq 8$. Introduce proof-only quantities

$$D = XY^2, \quad A = 4\big(1 + D(D+1)k^2\big), \quad b = 2L + k, \quad t = Z/U, \quad \rho = t + b.$$

The exact norm identity is

$$N/U^4 = (\rho^2 - A - G)^2 - 4AG. \tag{L1}$$

Because $D(D+1) \geq 0$ for integer D, $A \geq 4$. At $N = 0$, (L1) forces $G \geq 0$. Therefore $B_{\mathrm{ud}} = G + 1 + p_2^2 + p_3(p_3 + 1) > 0$, which excludes $Y = 0$ and $k = 0$. Directly, $B_{\mathrm{ud}} \leq Y^2 k^2 q^2$. Since $|D| \geq 8$, one has $D(D+1) \geq D^2/2$, and hence

$$A \geq 4 + 2X^2Y^4k^2 \geq 4 + 32q^2Y^4k^2 > Y^2k^2q^2 \geq B_{\mathrm{ud}} > G. \tag{L2}$$

In particular $\rho \neq 0$, since (L1) with $\rho = 0$ would force $A = G$. The rational numbers

$$p = \frac{\rho^2 - A + G}{2\rho}, \qquad u = \rho - p$$

satisfy $p^2 = G$ and $u^2 = A$. A rational square root of an integer is integral, so $p, u \in \mathbb{Z}$, and u is even. It follows that

$$t_{\mathrm{old}} = t - p = u - b \in \mathbb{Z}, \qquad t_{\mathrm{old}}^2 + 2(2L + k)t_{\mathrm{old}} + k^2 - 4 = 0. \tag{L3}$$

The argument explicitly handles equal square classes: for two roots, the separation $A > G$ rules out the sole cancelling-sum obstruction. It does not claim that pairwise distinctness suffices for a norm with more than two roots.

5.4 Complete maps in both directions

At any new zero, retain all shared coordinates and set

$$p_{1,\mathrm{old}} = p, \qquad n_{\mathrm{old}} = n - p.$$

Since Z is affine in n with coefficient U, (L3) makes the old first carrier zero. Its strong and auxiliary residuals are unchanged, and its guard is zero because $B_{\mathrm{ud}} = p^2 + 1 + p_2^2 + p_3(p_3 + 1)$. This is a literal zero of the pinned fifteen-variable polynomial, so its complete ordinary soundness theorem now applies. No premature CRT or canonical assumption has been used.

Conversely, from any old zero, delete p_1 and set $n_{\mathrm{new}} = n_{\mathrm{old}} + p_1$. Then $G = p_1^2$ and $Z_{\mathrm{new}} = Z_{\mathrm{old}} + Up_1$. The universal identity

$$N_{\mathrm{new}} = C_{1,\mathrm{old}}\big[C_{1,\mathrm{old}} + 4Up_1(Z_{\mathrm{old}} + Ub) + 4U^2p_1^2\big]$$

gives $N_{\mathrm{new}} = 0$, with the other two residuals unchanged. This applies, in particular, to every member of the authentic canonical source34 family supplied from a finite ZFC refutation. No new rank-tuning or two-square representation theorem is required.

5.5 Exact coefficient, cost and boundary

The degrees of C_1, B, C_2, G, N are $22, 12, 16, 18, 60$. For the leading homogeneous forms, write

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v, \quad R_0 = q_0^4 z - 12q_0^3 wC,$$
$$a_0 = X_0 Y_0, \quad U_0 = 8q_0^3 X_0 Y_0.$$

Then

$$N_{60} = U_0^4 R_0^4 q_0^8 (q_0^8 + 16X_0^2 Y_0^4),$$
$$B_{12} = -4a_0^2 r^2 c^4, \qquad (C_2)_{16} = 2a_0^2 f^4 R_0 y.$$

Their product is the unique degree-88 part of P. In particular,

$$[j^{56} w^8 v^8 z^5 r^2 c^4 f^4 y]P = -2^{87} 3^{64} \beta^{56} \neq 0.$$

Thus the degree is attained on the actual fixed coefficients. The 119-gate count includes all scalar multiplications and squares. This lowers the historical fourteen-variable degree-108/119-gate endpoint's degree at unchanged evaluator cost; it is a different tradeoff from the fifteen-variable degree-44/111-gate source.

The independent audit checks all 33 named registers, literal liveness and gate counts, the nonzero coefficient, complete coefficient bytes, source pins, identities, residue cases, signed bounds, and deliberate gate mutations. Normal and optimized runs pass. These finite checks do not replace the unbounded source34 compiler/refutation theorem, whose source closure remains an explicit dependency.

6 Parity separated roots with thirteen integers

This endpoint uses thirteen unrestricted integers, has exact degree 156, and has a division-free evaluator of 127 binary operations: 75 multiplications, 27 additions and 25 subtractions. It eliminates both the strong Pell root and the triangular positivity coordinate from the fixed fifteen-variable free-scale source. Its zero existence is equivalent to that source by maps on every zero, not just a canonical subclass. The same ordinary source34 compiler/calibration therefore supplies the ZFC-provable equivalence between zero nonexistence and syntactic Con(ZFC); this is not a proof of consistency or a new end-to-end formal-kernel certificate.

6.1 Complete polynomial

The unknowns are

$$j, C, z, h, r, v, w, T, c, n, y, p_1, p_2 \in \mathbb{Z}.$$

The six constants $\beta, K_0, M_C, M_F, A_c, B_c$ are the unchanged complete source payloads, with M_F already shifted. Define only polynomial abbreviations:

$$
\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \\
I = (2r + 1)c^2, \quad D_s = \Delta I^2, \quad F_s = 1 + D_s, \\
Q = \Delta D_s, \quad V = F_s(cT - R) - c, \\
M = 2q^3, \quad U = MH, \quad K_{\mathrm{lo}} = X(2X - 3c) - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(Y - q^3), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_{\mathrm{ud}} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2(4(Y\eta)^2 + (kq^3)^2), \\
G = 4(B_{\mathrm{ud}} - p_1^2 - p_2^2) - 3, \qquad D_t = 1 + L(L + k), \\
C_1 = (Z + Uk)^2 + 4U(LZ - U), \qquad C_2 = V^2 - (Q - 1)y(2V + y) - 1, \\
g_a = U^2G, \quad d_b = 4U^2D_t, \quad f_c = 16U^2F_s, \quad g_c = g_a + f_c, \\
J_N = (C_1 - g_c)^2 - 4(g_af_c + d_bg_c), \\
N = J_N^2 - 64g_ad_bf_c(C_1 + d_b).
\end{gathered}
$$

The single equation is

$$\boxed{P = (N + 1)(C_2 + 1) - 1 = 0.}$$

Sharing D_s avoids recomputing $F_s - 1$ and saves one subtraction. All powers are fixed natural powers; the literal DAG has no divisions or radicals.

6.2 Norm and exact conjunction

For proof only, put $b = 2L+k$, $R_N = Z+Ub$ and $\rho = R_N/U$. The first-carrier identity gives $R_N^2 = C_1+d_b$. The symmetric norm

$$F_8(S; g, d, f) = \left[(S - g - d - f)^2 - 4(gd + gf + df)\right]^2 - 64gdfS$$

satisfies the polynomial identity

$$F_8(x^2; g, d, f) = \prod_{\epsilon \in \{\pm 1\}^3} (x - \epsilon_1\sqrt{g} - \epsilon_2\sqrt{d} - \epsilon_3\sqrt{f}).$$

Substitution $S = C_1 + d_b$, $(g, d, f) = (g_a, d_b, f_c)$ gives exactly N. Equivalently,

$$\frac{N}{U^8} = \prod_{\epsilon \in \{\pm 1\}^3} (\rho - \epsilon_1\sqrt{G} - 2\epsilon_2\sqrt{D_t} - 4\epsilon_3\sqrt{F_s}). \tag{T1}$$

Since N is a square modulo 4, $N \neq -2$. The integer product $(N + 1)(C_2 + 1) = 1$ therefore forces both factors to be 1; it is equivalent to $N = C_2 = 0$ with no sign premise on either residual. This is a tailored use of the standard unit-product device, not a claimed new general conjunction theorem.

6.3 Unconditional signs and square-class separation

The authentic odd $\beta > 1$ makes q even and nonzero. Also H is odd and nonzero, so $U \neq 0$. Since $|W| \geq 4$, X is even with $|X| \geq 8$. Thus $a = (X+1)Y$ is either 0 or has $|a| \geq 7$, giving

$$\Delta = (a+2)^2 - 1 \geq 3, \qquad F_s = 1 + \Delta I^2 \geq 1.$$

Writing $D = XY^2$, we have even D and $D(D+1) \geq 0$, whence $D_t = 1 + D(D+1)k^2$ is a positive odd integer. Also $G \equiv 1 \pmod 4$. If $G < 0$, every factor in (T1) has a nonzero imaginary part from its sole imaginary radical, while $\rho, \sqrt{D_t}, \sqrt{F_s}$ are real. Hence at a zero $G > 0$; $G = 0$ is excluded by its residue.

Write the positive radicands uniquely as

$$G = u_1^2 d_1, \qquad D_t = u_2^2 d_2, \qquad F_s = u_3^2 d_3,$$

where the d_i are positive squarefree and the u_i are positive integers. The numbers u_1, u_2, d_1, d_2 are odd. A vanishing norm factor gives

$$\rho = \epsilon_1 u_1 \sqrt{d_1} + 2\epsilon_2 u_2 \sqrt{d_2} + 4\epsilon_3 u_3 \sqrt{d_3} \in \mathbb{Q}.$$

Distinct squarefree radicals, including $\sqrt{1}$, are linearly independent over $\mathbb{Q}$: sign-change automorphisms in the finite multiquadratic extension, averaged against a chosen character, isolate each coefficient. Group terms with equal squarefree part. A group containing the G term has an odd coefficient sum. A group containing D_t but not G has coefficient sum $2 \pmod 4$. A remaining group contains only the nonzero F_s term. Thus no irrational square-class group can cancel. An even squarefree class, in particular, could contain only F_s and is also impossible. Consequently all three radicals are integers. With the signs of the vanishing factor, write

$$g = \epsilon_1 \sqrt{G}, \qquad \tau = \epsilon_2 \sqrt{D_t}, \qquad f = \epsilon_3 \sqrt{F_s}; \qquad \rho = g + 2\tau + 4f.$$

Here g is odd; no oddness of F_s was assumed. This is an exact valuation argument within common square classes, not an inference from pairwise distinct radicands or finite tests.

6.4 Complete maps and canonical completeness

From any new zero restore the old coordinates by

$$f_{\text{old}} = f, \qquad p_{3,\text{old}} = (g-1)/2, \qquad n_{\text{old}} = n - g - 4f,$$

retaining every other coordinate. All three restored coordinates are integral. Affineness of Z gives $Z_{\text{old}}/U = 2\tau - b$, making its first carrier zero. The identity $f^2 = F_s = 1 + \Delta I^2$ makes its strong row zero; its auxiliary row is unchanged because the new F_s equals the old f^2. Finally,

$$p_{3,\text{old}}(p_{3,\text{old}} + 1) = (G-1)/4,$$

so its guard is precisely $B_{\text{ud}} = 1 + p_1^2 + p_2^2 + p_{3,\text{old}}(p_{3,\text{old}} + 1)$. This is a literal source zero, after which its ordinary soundness theorem applies. No preliminary CRT, positive orientation or canonical restriction was used.

Conversely, from any old zero delete f, p_3 and put

$$n_{\text{new}} = n_{\text{old}} + (2p_{3,\text{old}} + 1) + 4f_{\text{old}}.$$

The old rows imply $F_s = f_{\text{old}}^2$, $G = (2p_{3,\text{old}} + 1)^2$, and $Z_{\text{old}}/U + b = 2\tau$ with integral $\tau^2 = D_t$. The latter integrality is explicitly proved by the pinned source's monic carrier argument. One factor in (T1) is therefore zero and the auxiliary row is unchanged. This map applies to every old zero, in particular to the exact canonical source34 family supplied from a finite refutation. No extra canonical tuning or simultaneous representation theorem is required.

6.5 Exact coefficient and verification

The degrees of $C_1, F_s, D_t, G, Q, V, C_2, N$ are respectively $22, 12, 18, 18, 18, 17, 36, 120$. Put

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v, \quad R_0 = q_0^4 z - 12q_0^3 wC,$$
$$a_0 = X_0 Y_0, \quad U_0 = 8q_0^3 X_0 Y_0.$$

The leading forms are

$$N_{120} = 2^8 U_0^8 R_0^8 q_0^{16} (q_0^8 + 4X_0^2 Y_0^4)^2,$$
$$(C_2)_{36} = 32 a_0^6 r^4 c^8 R_0 y.$$

Their product is the unique degree-156 part, and

$$[j^{106} w^{14} v^{14} z^9 r^4 c^8 y] P = 2^{171} 3^{120} \beta^{106} > 0.$$

The exponent sum is 156, so this is actual degree attainment on the fixed coefficient slice. All thirteen variables and six coefficient ports are live; the unchanged numerals occupy 615883 uncompressed bytes.

The independent normal and optimized audits authenticate the source and numeral bytes, check every named register, multiply all eight signed-root factors, verify the centered norm schedule and lift, prove the coefficient and degree, check literal cost and liveness, and reject all live-gate mutations. Finite checks are not substituted for the square-class proof or the unchanged unbounded source34 compiler/refutation theorem. This endpoint is a new variable/degree/cost tradeoff, not a domination claim over the fourteen-variable degree-88 or fifteen-variable degree-44 constructions.

7 Canonical saturation of the doubled Pell product square

Two new literal polynomials in ten strictly positive integer unknowns have

$$(\text{operations}, \text{degree}) = (81, 750), \qquad (82, 560).$$

They combine canonical q-cube saturation with the incoming doubled-Pell product-square construction. Compared with the latter's $(84, 436)$ and $(85, 328)$ points, this saves three gates at a degree cost. The complete *directly indexed BFS943 compact44* coefficient tuple is unchanged. This is an ordinary arithmetic reduction using the retained source/native calibration, not a variable-count record, a global minimum, a consistency proof, or a newly generated formal proof endpoint.

7.1 Literal equations

The independent variables are $J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}$. The four fixed numerals β, K_0, M_C, M_F are supplied in the coefficient payload and satisfy $\beta \geq 31$ odd, $K_0 > 0$ and $0 < M_C < \beta < M_F < 2\beta$. Define

$$\begin{aligned}
q &= \beta J + 1, & h &= q - 1, & X &= wq, \\
\mu &= q^2 + 1, & Y &= q^3 s, & k &= \eta + \zeta, \\
c &= kY + \eta, & A &= \mu Y(X+1) + 1, & \Delta &= A^2 - 1, \\
H &= 4A - 5, & V_0 &= X\mu^2 Y^2, & U &= ic^2, \\
E &= 1 + V_0(V_0+1)k^2, & & & & \\
R &= [h(q-F) + \alpha](q^2-1) + (M_C + qM_F)J, & & & & \\
e &= k - R - 1, & C &= q - F - \alpha, & B &= (K_0 + w)C - F,
\end{aligned}$$

$$a = \Delta U^2 E, \qquad F_0 = 1 + a, \qquad G = EF_0,$$
$$F_{\mathrm{sq}} = 1 + 4aF_0, \qquad Q = \Delta(F_{\mathrm{sq}} - 1),$$
$$b = c(T + \alpha) - R, \qquad V = F_{\mathrm{sq}} b - c,$$
$$K_+ = X(3c - 2X) + 2, \qquad M = h\mu^2 H.$$

Choose $L = V^2$ for the 81-gate equation or $L = bV$ for the 82-gate equation, and put

$$Z = h\mu^2[H(n - LB) + K_+] - H[\mu^2 B + he],$$
$$R_G = M^2 G,$$
$$\boxed{N = Q(Z^2 - R_G V^2) + (R_G - Z^2) = 0.}$$

All symbols are polynomial abbreviations, not additional unknowns. With proof notation $\delta = Q - 1$, $D_y = QV^2 - 1$, $\kappa = K_+/H$, $\rho = e/\mu^2$, $t = B/h$ and $W = 1 + hL$,

$$N = \delta Z^2 - M^2 G D_y, \qquad \frac{Z}{M} = n + \kappa - Wt - \rho. \tag{1}$$

7.2 Normalization and positive soundness

For the matching source133 polynomial, the substitutions $Y_{\mathrm{old}} = q^3 s$ and $n_{\mathrm{old}} = n - s$ give whole-polynomial identities

$$Z_{\mathrm{old}} = q^3 Z, \qquad M_{\mathrm{old}} = q^3 M, \qquad N_{\mathrm{old}} = q^6 N.$$

The exponent is six, because this output is quadratic in the cleared coordinates. The shifted $n - s$ need not be positive, so this identity alone is not used as a positive soundness argument.

All source kernel inequalities hold because $Y > 0$. On every positive input, G, δ, D_y, M are positive. Pell doubling gives

$$F_{\mathrm{sq}} = (1 + 2a)^2, \qquad Q = 4\Delta^2 U^2 G, \qquad \gcd(G, \delta D_y) = 1.$$

At a zero, $\delta(Z/M)^2 = GD_y$, so prime valuations force $G = g^2$. Since $F_0 = 1 \pmod E$, its coprime factors are squares: $E = \tau^2$, $F_0 = f_0^2$, $g = \tau f_0$. Rational adjacent-Pell descent makes $y = |Z/M|/g$ an integer satisfying

$$QV^2 - (Q - 1)y^2 = 1, \qquad Z/M = \varepsilon g y, \quad \varepsilon \in \{-1, 1\}.$$

The doubled strong pair $f = 1 + 2a$, $I = 2Ug$ satisfies $f^2 - \Delta I^2 = 1$, $Q = (\Delta I)^2$ and $c^2 \mid I$. The retained all-index folding theorem and the first norm therefore give $c = \psi_A(p)$ for odd $p \geq 3$, $k \geq 8V_0 + 4$, $b > c$, $V > 0$, $K_+ > 0$, and the one-sided bound $\rho < b/\mu^2$. These deductions use neither the transport sign nor carrier divisibility.

For $L = V^2$, the inequalities $g < a$ and $V > (F_{\mathrm{sq}} - 1)b > ab > 3g$ imply $gy < V^2/2$. If $B \leq -1$, equation (1) gives, for either root sign,

$$n = \varepsilon g y - \kappa + Wt + \rho < V^2/2 - V^2 + V/\mu^2 < 0.$$

For $L = bV$, the retained fifth-auxiliary-index argument first gives $V > 8Q^2$ and $b > 4g$. Hence $gy < 3bV/8 < L/2$ and negative B would give

$$n < L/2 - L + b/\mu^2 < 0.$$

Both contradictions concern the actual positive n, not $n - s$. Thus $B \geq 0$ and $C > 0$.

The later arithmetic now follows in its original order: rank recovery gives $p = R$, then parity gives even q; only then denominator coprimality gives $\mu^2 \mid e$. The first-index comparison and ratio bounds

hold unchanged. Clearing the integral carrier modulo H gives $H \mid hK_+$ and therefore the source's weaker premise $H \mid q^3hK_+$. Its whole-product projection and height estimate force $X = 2^R$, without cancelling nonunits. Then κ is integral, and $W = 1 \pmod h$ gives $h \mid B$. Finally $q^3 \mid Y$ is literal. The fixed-minus parity, exact half-binomial rounding and fresh unscaled source reconstruction yield a finite refutation through the unchanged whole indexed calibration.

7.3 Canonical completeness

From a finite refutation, use the source133 canonical outer and first Pell data, including $q^3 \mid Y$. Choose its fresh seed period divisible by $c^2\tau$, double that seed, and use its explicit odd auxiliary witness. The source's positive carrier is

$$n_{\text{old}} = gy + Wt + \rho - \kappa - Y/q^3 > 0.$$

Set $s = Y/q^3 > 0$ and $n = n_{\text{old}} + s > 0$. The normalization identity gives $q^6N = N_{\text{old}} = 0$, hence $N = 0$. Equivalently $n = gy + Wt + \rho - \kappa > 0$, since $g > \kappa$ and $Wt, \rho \geq 0$. This includes native transport $t = 0$ and does not assume simultaneous solvability of unrelated Pell equations. Thus absence of a positive zero is equivalent to Con(ZFC) under exactly the retained ordinary calibration.

7.4 Exact degree and reproducible arithmetic

The degrees of $A, c, U, E, a, G, F_{\text{sq}}, Q, b, V, M, B$ are

$$8, 5, 11, 30, 68, 98, 136, 152, 6, 142, 13, 2.$$

For $L = V^2$, $\deg Z = 299$, so the two summands of $\delta Z^2 - M^2GD_y$ have degrees 750 and 560. For $L = bV$, their respective degrees are 478 and 560. Hence no highest-degree cancellation is possible. With $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $t_0 = T + \alpha$, the full attained homogeneous forms are

$$[N_{81}]_{750} = 2^{14}q_0^{506}w^{46}s^{108}i^{20}k_0^{64}t_0^4(q_0 - F - \alpha)^2,$$
$$[N_{82}]_{560} = -2^{10}q_0^{384}w^{34}s^{80}i^{14}k_0^{46}t_0^2.$$

In particular the coefficients at $J^{506}w^{46}s^{108}i^{20}\eta^{64}T^4F^2$ and $J^{384}w^{34}s^{80}i^{14}\eta^{46}T^2$ are respectively $2^{14}\beta^{506}$ and $-2^{10}\beta^{384}$, nonzero for the actual fixed numerals.

The exact operation splits are $(48, 19, 14)$ and $(49, 19, 14)$ for products, sums and differences. All vertices and input ports are live. Three gates are saved by adding $Y = q^3s$, deleting four old carrier gates, and changing the remaining carrier rows. The standard-library checker verifies all coefficient numeral bytes, complete graph bindings, normalization, full polynomial rays, local identities and actual gate mutations, both normally and under Python optimization. The proof, not these finite controls, supplies the unbounded equivalence. Three older gzip containers have different headers from source133, but their decompressed coefficient bytes match its pinned hashes exactly; this packaging distinction is recorded explicitly.

8 An eleven-integer single-product carrier at degree 648

The new nondominated point is an ordinary polynomial in eleven unrestricted integers, of exact degree 648, evaluated by 115 binary arithmetic operations. It improves the recovered 654/117 endpoint in both degree and cost. Three audited comparisons have degree/cost pairs 640/118, 580/119, and 574/122; they are dominated by other recovered equations and are not advertised as frontier advances. None replaces the retained 352/132 low-degree or 840/110 low-cost endpoint.

8.1 The complete equation

The variables are $j, C, z, h, r, v, w, T, c, n, s \in \mathbb{Z}$. All other symbols below are polynomial abbreviations. Use the same six authenticated fixed numerals $\beta, K_0, M_C, M_F, A_C, B_C$, with the already shifted M_F. Their complete bytes and reconstruction recipe accompany the literal DAG. For the primary weak-scale equation define

$$
\begin{aligned}
J &= 6j+1, & q &= \beta J+1, & W &= 12w+4,\\
X &= qW, & Y &= qv, & E &= XY,\\
a &= E+Y, & A &= a+2, & H &= 4a+3,\\
\Delta &= a^2+H, & F &= (K_0+W)C - z(q-1),\\
R &= (q(q-F)-C)(q^2-1)+(M_C+qM_F)J,\\
k &= R+1+hE, & \eta &= c-Yk,\\
D_{\mathrm{err}} &= (A_CF-q)^2+(B_CC-q)^2,\\
B_0 &= k^2(q^2-D_{\mathrm{err}})-4q^2\eta^2, & B &= B_0,\\
\ell &= XY^2, & L &= \ell k, & D_t &= 1+L(L+k),\\
D &= 3B-4, & D_g &= 1+4DD_ts^2, & P &= D_tD_g,\\
D_m &= 1+\Delta c^2, & B_P &= D_mP, & U &= Arc^2,\\
D_f &= 1+\Delta U^2B_P, & Q &= \Delta^2U^2B_P, & \delta &= Q-1,\\
V &= D_f(c-R+4UB_PT)-c, & C_y &= QV^2-1,\\
K &= X(3c-2X)+2, & M &= 2q^2, & D_c &= HM,\\
z_c &= D_cT+MK+H(v-q^2),\\
A_c &= (z_c^2+3D_c^2)(z_c^3-19D_c^3),\\
S &= srHM^5, & Z &= Sn-A_c.
\end{aligned}
$$

The final polynomial is simply

$$\boxed{\mathcal{F} = \delta Z^2 - S^2B_PD_fC_y.}$$

There are no divisions, radical gates, variable exponents or unnamed witnesses. All scalar products and powers are charged in the literal DAG. For the primary direct equation put $\lambda = 0$. The simple center replaces Z by $S(n+\lambda V)-A_c$, where $\lambda = AUB_P$. The free-scale comparisons instead take $Y = v$, $B = Y^2B_0 - k^2q^8$, $M = 2q^3$, and replace $H(v-q^2)$ in z_c by $H(v-q^3)$.

8.2 Soundness, including the early zero-scale branch

Unconditionally, the authentic front gives q even, $q \equiv 2 \pmod 3$, $8 \mid X$, $X \equiv 2 \pmod 3$, and $|X| \geq 4|q|$. Also H is nonzero odd, $\Delta > 1$, $D_m \geq 1$, and $D_t \geq 1$. The odd integers D_g, P are nonzero. The integers D_f, δ, C_y cannot vanish. As $\pi(t) = (t^2+3)(t^3-19)$ has no rational root, $A_c = D_c^5\pi(z_c/D_c) \neq 0$. Therefore a zero of $\mathcal{F}$ has $S \neq 0$, hence $r, s \neq 0$.

Put $D_a = \delta C_y$. The three nonzero factors B_P, D_f, D_a are pairwise coprime: the last two are 1 modulo B_P, while

$$Q \equiv -\Delta, \quad \delta \equiv -A^2, \quad V \equiv -c, \quad C_y \equiv -D_m, \quad D_a \equiv A^2D_m \pmod{D_f}.$$

Since $D_f \equiv 1 \pmod A$ and $\pmod{D_m}$, the claim follows. The equation gives

$$(\delta Z/S)^2 = B_PD_fD_a.$$

Thus each absolute factor is a square. Since ΔA^2 is divisible by four, $D_f \equiv 1 \pmod 4$, so it cannot be a negative square. For $c \neq 0$, negativity of B_P would make $D_f < 0$; for $c = 0$, the equation directly gives $Z^2 = S^2 B_P$. Hence $B_P, D_f, D_a > 0$. Write their positive roots as p_0, f, e. Then $P > 0$ forces $B \geq 2$, yielding the centered strip and authentic packing bounds. The free guard additionally forces $|Y| > |q|^3$.

A crucial order-of-proof point is that carrier integrality comes *before* main-root folding. If $c = 0$, it follows immediately from $Z^2 = S^2 B_P$. Otherwise set $I = Up_0$ and $T_o = p_0 T$. Then $Q = (\Delta I)^2 > 1$, and the inherited adjacent-Pell denominator descent makes $y = e/\delta$ integral. Thus $Z^2 = S^2 p_0^2 f^2 y^2$, so in either branch $S \mid A_c$. This avoids assuming $|A| \geq 3$ while the weak scale might still be zero.

The valuation of the monic leading term in

$$A_c = z_c^5 + 3D_c^2 z_c^3 - 19D_c^3 z_c^2 - 57D_c^5$$

then proves $M \mid z_c$, and hence $M \mid H(v - q^2)$ in the weak case. Since $H \equiv 3 \pmod q$, $q \not\equiv 0 \pmod 3$, and H is odd, $\gcd(H, M) = 1$. Consequently

$$v = (2v_o + 1)q^2, \qquad Y = (2v_o + 1)q^3.$$

Also $H \mid K^5$. The restored magnitude and centered strip exclude $c = 0$. Now the inherited early absolute-folding argument applies to the strong and auxiliary norms; it yields $D_m = d^2$ without using a first Pell root. From $p_0^2 = D_m P$ it follows that P is a square, and $\gcd(D_t, D_g) = 1$ splits it into $D_t = \tau^2, D_g = g^2$.

All pre-projection signed-rank hypotheses are now present, including $c^2 \mid I$, $V \equiv -c \pmod f$, $V \equiv -R \pmod c$, and $k = R + 1 + hXY$. The retained nonodd strong-rank and first-index arguments give $R = 2t + 1$, $t \geq 36$, and $|H| > |X|(|X| - 3)^t$. Writing $\sigma = \operatorname{sign}(c)$,

$$\Pi = (\sigma X - 2^R)(2^R \sigma X + 1),$$

the signed projection gives $H \mid \Pi^5$. The free-scale version first gives only $H \mid q^{15}\Pi^5$; its early carrier nevertheless gives Y even. Both are covered by the stronger bound, with $x = |X|$,

$$|q|^{15}|\Pi|^5 < 2^{20t-10} x^{25} < x^{t+1}/2^t < |H|,$$

because $x \geq 2^{64}$ and $64(t - 24) - (21t - 10) = 43t - 1526 \geq 22$. Thus $\sigma X = 2^R$; residues modulo three force $\sigma = 1$. Only after this projection does the free case cancel its scale modulus, since $q \mid 2^R$ and H is odd.

The inherited surviving signs and uniform centered half-binomial theorem then give $c, k, Y > 0$ and $0 < \eta/k < 1/4$. Fresh strong/auxiliary private witnesses restore the literal old signed source zero, as detailed in `PROOF.md`; no oddness of the incoming I/c^2 is assumed. Only this old zero is passed to the retained native/compiler/refutation calibration.

8.3 Completeness and its quantifier order

A finite syntactic refutation supplies the canonical outer tuple with $D_m = d^2$, $D_t = \tau^2$, A positive even, $R \equiv c \equiv 3 \pmod 4$, and $B > 64$. Since $D_t \equiv 1 \pmod 3$, $(3B - 4)D_t$ is positive and $2 \pmod 3$, hence nonsquare. Fix a positive Pell guard pair s, g, and put $p_0 = d\tau g$. Choose one strong pair I, f with $4sAp_0c^2 \mid I$, and set $r = I/(Ap_0c^2)$. Keep it fixed. Every prime of $p_0 sr$ divides I, so the retained supported-modulus interpolation theorem realizes every T_o residue modulo $p_0 sr$ for this same pair.

The canonical projector gives $\kappa = K/H \in \mathbb{Z}$, and

$$A_c/S = H^4 \pi(T + \kappa + v_o)/(sr).$$

Choose an intersective root a_0 of $\pi(T + \kappa + v_o)$ modulo sr, prescribe $T_o \equiv p_0 a_0 \pmod{p_0 sr}$, and use supported interpolation. Then $T = T_o/p_0$ and A_c/S are integers. Set $n = A_c/S + p_0 fy - \lambda V$. The equation vanishes. The guard and strong pair are never reselected after choosing the residue.

8.4 Exact degree and replay boundary

For the weak direct equation the register degrees are

$$\deg(B_P, U, D_f, Q, V, C_y, S, A_c) = (72, 7, 94, 102, 174, 450, 16, 35).$$

The unique dominating product $S^2 B_P D_f C_y$ has degree 648. The attained monomial

$$j^{372} w^{64} v^{106} s^{16} r^{12} c^{34} z^{42} T^2$$

has coefficient $2^{546} 3^{443} \beta^{372} > 0$. For the simple center, $\lambda^2 - B_P D_f = U^2 B_P^2 - B_P$; exact expansion leaves a unique leading term $S^2 Q U^2 B_P^2 V^2$, of degree 640. The free analogues have degrees 580 and 574.

The independent standard-library checker does not import the builder. It binds every live arithmetic gate in free polynomial rings, checks all identities and exact leading monomials, reconstructs every fixed numeral, authenticates retained sources, and rejects every single-gate mutation. Normal and optimized Python runs agree. These finite checks are not substituted for unbounded Pell, interpolation, or compiler theorems. The result is an ordinary ZFC-provable equivalence between universal nonvanishing and syntactic Con(ZFC), not a proof of consistency or a new end-to-end formal certificate.

9 What is checked and how to reproduce it

9.1 Independent mathematical and literal reviews

The direct review checks all eight retained charts, not only the headline three. It independently reconstructs the raw signed-domain modifications, all 18 power lowerings, 15 primary norms, the complete frame and cap pool, 11 successor residue ideals, private graph scopes, Horn identities, monic eliminations, translations and every collected quartic coefficient. Eight deliberately corrupted artifacts must be rejected. The separate ordinary audit treats the negative comparison-base branches, reconstructs private helper witnesses and gives the self-radix soundness and completeness order. Finite helper examples are diagnostic controls, not the proof.

The independent compact audits certify all live gates by exact local polynomial identities, recover full leading forms or specified nonzero leading monomials, authenticate source and coefficient bytes and examine the unbounded witness maps. Across the four eleven-integer circuits and the two low-degree circuits they reject 720 individual live-gate corruptions in each Python mode. The positive review rejects a further 163 individual gate corruptions, as well as targeted scaling, shear and operand changes. Thus 883 single-gate corruptions are rejected per mode; this count excludes the many other algebraic checks and supplemental controls.

In particular, the positive review reconstructs the complete indexed BFS943 coefficient tuple from its printed finite decoder and verifies the $N_{\mathrm{old}} = q^6 N$ normalization by exact symbolic identities. The integer review independently reconstructs the source34 recipe and verifies the interpolation chronology, rather than using numerical evidence for an unbounded statement. The root ordinary review separately checks the thirteen-integer square-class argument, literal old-zero maps, unit-product conjunction and attained coefficient.

9.2 Finite offline replay

The exact receipt environment is Python 3.12.14 with SymPy 1.14.0. The direct arithmetic and the compact producer builders use the standard library; selected independent symbolic audits use SymPy. All equations have literal integer coefficients and fixed exponent schedules. The finite coefficient recipes construct fixed data and do not simulate the encoded machine or search for a refutation.

From the extracted release root, run

```
python -B reproduce.py --receipt ../round17-replay/REPRODUCE_CHECK.json
```

The runner first checks the release inventory and reviewed code hashes. It then makes separate temporary copies, deletes the declared generated outputs, rebuilds and checks them under normal and optimized Python, and compares exact bytes with the sealed references. Environment-bearing receipts are matched to their own mode. Changed and missing equations, coefficient corruption, unsafe paths and a corrupted ZIP member are explicit negative controls. The original release is not modified.

To verify a delivered archive without rerunning algebra, add its path:

```
python -B reproduce.py --archive ../release.zip --verify-only
```

The ZIP builder uses sorted portable member names and fixed metadata, builds two independent archives, requires byte identity, and checks CRC, complete inventory and every member byte. The replay summary also binds executed source files and the exact replay plan, so a later code change cannot silently reuse an old replay result. Integrity manifests are not signatures or mathematical proofs.

9.3 The report and remaining boundary

The modular editable source is `report/report.tex`. Its copied producer sections have exact source and destination hashes, and every presentation replacement is recorded. The root result data generate the summary table; an independent report-data checker recollects the three headline quartics, counts the compact graphs and authenticates all fixed payloads. The self-contained TeX attachment expands only local source inclusions and retains a separate source map.

Run `report/build_report.sh` to rebuild the modular report outside the sealed tree. Shell escape is disabled. On a minimal Debian installation without generated TeX caches, the script creates a temporary format from already installed source files; it does not change the system installation. The delivered PDF's page-by-page inspection is recorded in `report/QA_RECEIPT.json`. PDF bytes may differ across TeX installations; the deterministic arithmetic and ZIP claims do not imply universal PDF-byte reproducibility.

The mathematical boundary remains explicit: ordinary reductions compose with retained compiler, Pell, calculus and refutation-calibration theorems. Neither these finite programs nor a correctly rendered PDF constitute a new end-to-end proof-assistant certificate, a consistency decision, or a global optimality claim. No repository write, commit or third-party publication was performed.

10 Provenance and source guide

1. Vladimir Reshetnikov, *ProveIt*, the pinned snapshot and later explicitly identified comparison updates in `baseline/FRONTIER.md`. The download ledger records exact immutable GitHub URLs, Git blob identities, SHA-256 hashes and sizes. The finite comparison audit re-expands the selected quartics without executing downloaded builders.

2. *Signed Pell caps and free scale carriers*, the preceding round 16 delivery of 25 October 2026. Its complete extracted source is retained at `prior/wave16/`. Its original manifest has SHA-256

   ```
   cc064fadb4be95d4a16576978b5397909
   ccdac62e532b412722634327085a899.
   ```

   Every one of its 1,592 recorded file entries has been verified.

3. `direct/PROOF.md` and its source ledger retain the signed-cap, fused-ninth-factor, support-product, cofinal and self-radix interfaces. The independent ordinary proof audit and literal coefficient checker are in `review/direct/`. The three centered endpoints each have complete residual and collected coefficient files.

4. `lowdegree/` contains the literal low-degree equations, ordinary proofs, source pins and exact coefficient payloads. The independent norm review is retained under `review/lowdegree/`.

5. `integer/PROOF.md` states the eleven-integer carrier's complete soundness, interpolation order and canonical converse, with all immediate unbounded theorem statements copied and pinned. Its three dominated siblings are retained as comparisons rather than counted as frontier improvements. The independent review is under `review/integer/`.

6. `positive/PROOF.md` and `positive/source/` preserve the source 133 product-square theorem and exact indexed-native coefficient calibration. In particular, `positive/source/provenance/indexed_constant_specification.tex` contains the complete finite constant recipe, and `positive/source/provenance/indexed_native_proof.md` contains the matching native theorem. The exact algebra and full coefficient reconstruction review is in `review/positive/`.

Licenses and notices remain with their inherited source closures. The root `DEPENDENCIES.md` distinguishes the current finite execution closure from historical evidence and unbounded ordinary assumptions. Recorded old working-source labels are resolved by `SOURCE_LOCATION_MAP.json`; the pinned source texts themselves are not silently edited.